Structure-based druggability of CA2 and CR2 in colon adenocarcinoma

BIOT 6900 · Module 4 Assignment · Drashti Bharat Bhanushali · 22 October 2026

I took my two Module 3 targets forward: CA2 (P00918, rank 1 in Module 2) and CR2 (P20023, rank 11). Module 3 showed that both look like proteins that have been drugged before. In this module I wanted to see whether their predicted structures actually have somewhere a drug could bind, and how much I can trust those structures.

Anchor check: p53 (Step 8)

In lab Part 2, the AlphaFold model of p53 matched chain B of 1TUP with an RMSD of 0.68 Å over 192 confident residues, so the DNA-binding domain is predicted very well. Of the Module 3 hotspots, C176, H179, C238 and C242 are zinc ligands (2.1- 2.3 Å from the zinc), R248 and R273 contact DNA (3.4 and 2.5 Å), R249 is near DNA, and R175, Y220, G245 and R282 are structural/other.

This lines up with my Module 3 scores. Positions 176, 238 and 242 were among the least tolerated in the whole protein (mean scores -13.34, -11.81 and -11.10), and R248Q was the most extreme hotspot variant (-9.55, 5.2nd percentile). So from sequence alone, the language model picked out three of the four zinc ligands and a key DNA contact. Y220C is the exception. It had the mildest hotspot score (-3.80, 33.4th percentile), and Y220 neither holds zinc nor touches DNA, but the mutation opens a druggable crevice. A sequence score can't see that pocket, and neither can an AlphaFold model of normal p53, because the pocket only exists in the mutant.

1. Which parts can I trust?

	CA2	CR2
Residues	260	1,033
Mean pLDDT	97.4	74.9
% above 90 / % below 50	98.5 / 0.4	1.3 / 6.3
Stretches of 10+ below 50	none	1-19, 1003 -1033
Confident domains	1 (4-260)	8
Mean PAE between domains 1 and 2	-	26.3 Å

CA2: I trust the whole structure. Nearly every residue is above 90, there are no disordered stretches, and the domain finder gives one domain from 4 to 260. That makes sense for CA2, which is a small, soluble, rigid enzyme that works on its own.

CR2: I trust the individual domains, but not how they are arranged relative to each other. Only 1.3% of residues are above 90 and the mean is 74.9, so the model is much less certain overall than for CA2. Both disordered stretches fit what is known about the protein. Residues 1-19 are the signal peptide (UniProt 1-20), which is removed before CR2 reaches the cell surface, and 1003-1033 is the cytoplasmic tail, which has no fixed shape on its own.

The domain finder split CR2 into eight blocks: 22-147, 154-342, 350-467, 499-594, 601-657, 689-781, 788-845 and 850-970. These follow CR2's 15 Sushi (SCR) domains in UniProt; for example, block 1 is Sushi 1-2 and block 8 is Sushi 14-15. The extracellular part of CR2 is a chain of small domains joined by short linkers, like beads on a string. The mean PAE between blocks 1 and 2 is 26.3 Å, which means AlphaFold is basically guessing how those two blocks sit relative to each other. So I would only trust

pockets or surfaces inside a single block, and not anything that depends on two blocks or on the overall shape of the protein. The transmembrane helix (972-999) is not part of any confident block. That region normally sits in a membrane, which is not in the model.

2. How well does each prediction match experiment?

CA2 vs 3HS4 chain A: almost an exact match. All 257 aligned pairs were identical and confident, the RMSD was 0.37 Å, and no residue was more than 2 Å off. The residues that moved most were 36 (1.3 Å), 128 (1.3 Å), 102 (1.2 Å), 129 (1.2 Å), 131 (1.0 Å) and 130 (1.0 Å). Residues 128-131 form one loop, and residue 130 (Phe131 in the crystal's numbering) is 3.4 Å from the bound acetazolamide. So the biggest difference is right next to the drug, which is in the crystal but not in the prediction. My guess is that the drug holds that loop in place and that this is not a prediction error, but at 1.0-1.3 Å the shift is small either way.

CR2 vs 1LY2 chain A: the notebook's RMSD can't be used. It reported 130 aligned pairs, 115 identical and confident, an RMSD of 38.45 Å, and 2.20 Å over 80 residues for domain 1. The 38.45 Å comes from the alignment, not the structure. 1LY2 only covers Sushi 1-2 (UniProt 20-149, from its DBREF record), while the model has 1,033 residues. The provided aligner has no extra cost for opening a gap, so breaking the short crystal sequence into scattered pieces costs no more than keeping it together. Because CR2's Sushi domains repeat, the aligner scored higher by matching single residues spread across the whole protein. 48 of the 130 pairs were outside 20-149, and the residues listed as differing most (440, 421, 433, 410, 383, 403) are not in the crystal at all.

3. Druggability calls

Rules: reject if the mean pLDDT of the lining residues is below 70 and specifically suspect if the PAE among them is above 10 Å or otherwise keep.

Target	Pocket	Rank	Score	Volume (Å³)	Mean pLDDT	PAE within (Å)	Verdict
CA2	6	1	0.317	315.7	98.4	1.2	keep
CA2	2	2	0.298	478.0	97.9	1.4	keep
CA2	9	3	0.126	225.6	98.6	0.9	keep
CA2	1	4	0.011	504.8	98.1	1.2	keep
CA2	12	5	0.011	228.9	98.5	0.9	keep
CR2	50	1	0.898	560.3	65.2	7.7	reject
CR2	42	2	0.006	219.6	59.2	3.8	reject
CR2	27	3	0.005	527.9	75.5	4.4	keep
CR2	34	4	0.005	437.4	66.5	7.2	reject
CR2	5	5	0.004	125.1	81.9	3.4	keep

CA2: "Structure supports a small-molecule pocket." All five top pockets pass both confidence checks, the model matches the crystal at 0.37 Å, and the real drug site shows up as a kept pocket (pocket 2; see section 6). Carbonic anhydrase inhibitors like acetazolamide are already approved drugs, so this is the best case: a rigid, confident domain with a pocket that is already formed.

CR2: "No confident pocket: consider another modality." The only pocket with a real score, pocket 50 (0.898), is rejected because the mean pLDDT of its lining residues is 65.2. Its lining residues (440-515) run from Sushi 7 into Sushi 8, across the gap between blocks 3 and 4 (468-498). The PAE among them is 7.7 Å, which is under the 10 Å cut-off, so the rejection comes from the low pLDDT and not from PAE. Three of the top five pockets are rejected, and the best kept one scores only 0.005.

For CR2 I would go with an antibody. Almost all of CR2 (21-971) is outside the cell, so an antibody can reach it. Each Sushi domain is confidently predicted on its own, so a single domain is a reasonable epitope, and an antibody only needs a surface to bind, not a cavity. This agrees with Module 3, where CR2's antibody prior (0.663) was much higher than its small-molecule prior (0.142), and its nearest drugged neighbours were the cytokine receptors CSF2RA, IFNGR2 and IL1RL2. I would avoid epitopes that span two domains, for the PAE reason above.

4. Structure vs my Module 2 ranking and Module 3 prior

The structure results agree with Module 3 for both targets. CA2 had my highest small-molecule prior (0.770), and its structure has a confident pocket where a known drug binds. CR2 was my one clear case of antibody prior over small-molecule prior, and its structure has no confident small-molecule pocket. Module 3 suggested this from sequence, and Module 4 backs it up with structure.

Module 2 is where it gets uncomfortable. Both genes are lower in tumour at the RNA and protein level (CA2: RNA −5.20, protein −2.16; CR2: RNA −1.97, protein −1.42). A good pocket doesn't help much if the plan is to inhibit a protein the tumour has already lost.

So I would keep both, but not as targets to inhibit. CA2 stays as a structurally confirmed protein that the tumour has lost, which makes it more useful as a differentiation marker than as a drug target. CR2 stays only as an antibody target, for example to reach the cells that still have it on their surface. If the goal is a protein to inhibit in tumour cells, I would replace both with genes that go up in tumour, such as MMP11 and TMEM97 from my Module 3 check. Structure tells me whether I can bind a protein, not whether I should.

5. Limits that apply to my targets

One snapshot, no metal. CA2's drug binds the zinc, but there is no zinc in the AlphaFold model. It didn't matter here because the pocket is already formed without it, but the model alone can't show how a drug binds the metal. To fix this, I would predict the complex with the zinc and acetazolamide (for example with Boltz-2 on the cluster) and check whether it reproduces the 3HS4 pose.

No partners, membrane or sugars. CR2 binds C3d through Sushi 1-2, sits in the membrane and is glycosylated (1LY2 has two NAG sugars). None of this is in the model, so an epitope I pick could be blocked by a partner or a glycan in real cells. To fix this, I would compare with the CR2-C3d complexes in the PDB (1GHQ, 3OED), predict the complex, and map the glycosylation sites before choosing an epitope.

Domain arrangement with no experimental check. The PAE says the arrangement of CR2's domains is uncertain, and the crystal only covers Sushi 1-2. To fix this, I would compare with the solution-scattering model of the full extracellular region (2GSX) or new cryo-EM or SAXS data, and redo the per-domain comparisons with a better alignment.

No pockets that only open on binding or mutation. Like Y220C in p53, a pocket that only forms in another conformation would not show up here. This matters most for CR2, where I found nothing. Molecular dynamics or a fragment screen on single Sushi domains could test this.

I also ran into limits in the tools. The provided aligner broke on a short crystal against a long protein. fpocket's volumes also changed a little between my two runs (CA2 pocket 6 was 320.3 Å³ the first time and 315.7 Å³ the second), while the scores, pLDDT, PAE and verdicts stayed the same, so I wouldn't read anything into small volume differences.

6. Ligand site

CA2 (3HS4). The crystal has three acetazolamide copies. AZM A701 is the functional one: it is 1.9 Å from the zinc, while A702 and A703 are 17.3 and 16.6 Å away, far from the active site. A701's site has

13 residues (92, 94, 96, 106, 119, 121, 130, 142, 196-199 and 208 in UniProt numbering), including the three zinc ligands His94, His96 and His119.

My best kept pocket is not where the drug sits. Pocket 6 (score 0.317) covers 0% of the A701 site; it is a small 8-residue pocket somewhere else. The drug site is pocket 2, which is ranked second (score 0.298), kept, and covers 85% of the A701 site. For the other copies, pocket 10 (rank 6, score 0.006) covers 67% of A702, and pocket 7 (rank 11, score 0.001) covers 90% of A703.

I think fpocket ranked the real site second because of its shape. The CA2 active site is a wide, cone-shaped opening with the zinc at the bottom, held by three histidines, and one of its walls is polar. It is open to solvent and fairly large (478 Å³). fpocket's score tends to favour smaller, enclosed, hydrophobic pockets, and this site is none of those. The two scores are close (0.317 vs 0.298), so it isn't a big miss, but it shows I should check the top few pockets against what is already known instead of trusting rank 1.

CR2 (1LY2). There is no drug-like ligand, only the two NAG sugars. The notebook treated them as ligands, but because of the alignment problem in section 2, their contact residues were mapped to the wrong places in the model (for example 440, 638, 863 and 998, which are not in the crystal). I am not drawing any conclusion from that output.